

Key Exam II

a) $f(z)$ is even. Therefore $a_{2k+1} = 0 \quad \forall k \in \mathbb{Z}$. Specifically $a_{-1} = 0$

$$b) \quad f(z) = (1 + 2z - z^2 + 2z^3 - z^4 + 2z^5 + z^6) \left(1 - \frac{z}{z^2} + \frac{1}{2} \frac{4}{z^4} - \frac{1}{6} \frac{8}{z^6} + \dots \right)$$

$$= z^6 + 2z^5 - 3z^4 - 2z^3 + 3z^2 + 2z + \frac{7}{3} - \frac{\frac{8}{3}}{z} + \dots$$

$$a_{-1} = -\frac{8}{3}$$

2a) by Thm Pg 2 on Int. Topics
$$\underline{I} = \operatorname{Re} \left\{ \pi i \left(\operatorname{Res}(f, i) + \operatorname{Res}(f, 2i) \right) \right\}$$

where $f(z) = \frac{e^{iz}}{(z^2+1)(z^2+4)}$

$$\operatorname{Res}(f, i) = \left. \frac{e^{iz}}{2z} \right|_{z=i}$$

$$\operatorname{Res}(f, 2i) = \left. \frac{e^{iz}}{2z} \right|_{z=2i}$$

$$\therefore \underline{I} = \pi \left(\frac{e^{-2}}{6} - \frac{e^{-4}}{12} \right)$$

b) by Thm Pg 3 on Int. Topics
$$\underline{I} = \operatorname{Im} \left\{ \pi i \operatorname{Res}(f, i) + \frac{\pi}{2} i \operatorname{Res}(f, 0) \right\}$$

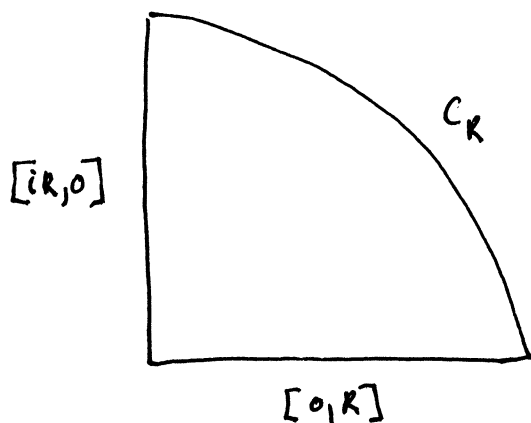
where $f(z) = \frac{e^{iz}}{z(z^2+1)^2}$

$$\operatorname{Res}(f, i) = \left. \left(\frac{e^{iz}}{z(z-i)^2} \right)' \right|_{z=i}$$

$$\operatorname{Res}(f, 0) = \left. \frac{e^{iz}}{(z^2+1)^2} \right|_{z=0}$$

$$\therefore \underline{I} = \pi \left(\frac{1}{2} - e^{-2} \right)$$

3. Let $f(z) = z^4 + z^3 - 2z^2 + 2z + 4$. Consider contour



On $[0, R]$ $z = x + i0$

$$\left. \begin{array}{l} \text{on } [0, 1] \quad f(x) \geq x^4 + x^3 + 4 \geq 4 \\ \text{on } [1, R] \quad f(x) \geq 2x + 4 \geq 4 \end{array} \right\} \arg f(z) = 0$$

On C_R $\Delta \arg f(z) \approx 2\pi$

On $[iR, 0]$ $z = 0 + iy$

$$f(iy) = y^4 + 2y + 4 + i(y^3 - y)$$

$$\Rightarrow f([iR, 0]) \text{ lies in RHP for all } z = iy$$

$$\therefore \Delta_{[0, R] \cup C_R \cup [iR, 0]} \arg f(z) = 2\pi$$

$$\Rightarrow f \text{ has 1 root in } \mathbb{Q}_I.$$

4. Consider $b_R(z) = \frac{z}{2R - z}$. $b : \{Rz < R\} \rightarrow |z| < 1$

Consider $g(z) = b_A \circ f(rz) : \mathbb{D} \rightarrow \mathbb{D}$, $g(0) = 0$

by Schwarz's Lemma

$$|g(z)| \leq |z|$$

$$\text{or } \left| \frac{f(rz)}{2A - f(rz)} \right| \leq |z|$$

$$\Rightarrow |f(rz)| \leq \frac{|z| 2A}{1 - |z|} \quad \text{for } z \in \mathbb{D}$$

$$\text{or } |f(z)| \leq \frac{2A |z|}{r - |z|} \quad \text{for } z \in B(0, r)$$

5. Let $f \in \mathcal{F}$. Then $f < 2i \frac{1+z}{1-z}$. Hence

$$|f'(0)| \leq 4. \quad \text{Since } 2i \frac{1+z}{1-z} \in \mathcal{F}. \quad \alpha = 4$$

6. Since $u \in \text{Harm}(\mathbb{C})$ and $\mathbb{C}$ is simply connected $\exists$
 $v \in \text{Harm}(\mathbb{C}) \rightarrow f = u + iv \in \mathcal{H}(\mathbb{C})$. Let $g = e^{-f}$.

Then g is entire and $|g(z)| \leq e^{-u} \leq e^2$. By
Liouville's Thm g is constant. Since f cont. $\Rightarrow f$ const.
 $\Rightarrow u$ constant

7a) If u and v are harmonic conjugates, then $f = u + iv$ is analytic
hence $g = f^2$ is analytic. $\text{Im } g$ is therefore harmonic,
 $\text{Im } g = 2uv \Rightarrow uv$ is harmonic

b) $u(x,y) = x$ is harmonic $v(x,y) = y$ is harmonic.
 $u \cdot v = xy$ is not harmonic

8. Let $A = \{z \in G \mid u(z) = u(z_0)\}$. A is closed because u is cont. on G .
Show A is open using same proof as given in Conway p. 253-254.
Since G connected, $A \neq \emptyset$, then $A = G$ and u is constant.

9. Let $g_n = f - t_n$. It suffices to show for any $K \subset \subset G$ that $g_n \rightarrow 0$ uniformly on K . Let $\varepsilon > 0$. Let $K_n = \{z \in K : g_n(z) \geq \varepsilon\}$. We have $K_n \supset K_{n+1}$ from the monotonicity of $\{g_n\}$. Each K_n is compact (since each K_n is a closed subset of K). Since for each $z \in K$ $\lim_{n \rightarrow \infty} g_n(z) = 0$, we have $\bigcap_{n=1}^{\infty} K_n = \emptyset$. The compactness of K implies $\exists N \ni K_N = \emptyset$. Hence, $\{g_n\} \rightarrow 0$ uniformly on K .